TripVote: Product Requirements Document

Section 1: Overview

Planning a group trip usually means scattered chats, polls, and spreadsheets. TripVote gives a group one place to agree on dates and activities, keep a shared itinerary, and compare budget to actual spending.

Section 2: Roles

Every trip has three roles:

- **Organizer:** creates the trip. Can do everything a Member can, plus the Organizer-only actions below.
- **Member:** joins the trip. Can propose options or recommendations, vote, and add/edit expenses.
- **Viewer:** can view the trip name and member list. A Viewer can accept an invitation to join a trip and has no stored record until they join.

Action	Organizer	Member	Viewer
See trip name and who's in it	✓	✓	✓
Share the trip link	✓	✓	
Accept an invitation or request to join			✓
View or add itinerary, availability, votes, expenses	✓	✓	
Approve or deny join requests	✓		
Remove members, set budget, edit or delete trip	✓		
Leave the trip		✓	

Section 3: User stories

A user is anyone with a TripVote account.

Story	Statement
US1	As a new user, I want to sign up and log in so my trips and votes are tied to me.
US2	As a user, I want to create a trip so my group has one shared home for planning.
US3	As an Organizer, I want to invite or approve users (Viewers) so that I control who can join us.
US4	As a Member, I want to add activities to a shared itinerary and vote on them, with changes appearing live, so we can decide together, plan ahead, and reserve them.
US5	As a Member, I want to mark which dates I'm free so the group can find dates that work.
US6	As a Member, I want to track spending against a budget so the trip stays affordable and fair.

Section 4: Non-functional requirements

Passwords are hashed and all inputs are validated. The layout works on both desktop and mobile screens. Basic features can be used with a keyboard.

Section 5: UI and mockups

Mockup (Claude Artifact, clickable prototype with fake data):
https://claude.ai/artifact/VpeXpJRxvQ86tuiQuUA84L

Key screens:

1. **Log in / Sign up (US1):** tab switch between Log in and Sign up. Sign up asks for name, email and password and shows the duplicate-email error. Submitting goes to the Dashboard.
2. **Dashboard (US2):** grid of the user's trips; each card shows name, role badge (Organizer / Member), destination, dates, member and activity counts. "New trip" opens an inline form (name, destination, start, end) with the "end date can't be before the start date" error. Pending invitations appear in a banner.
3. **Invite link as Viewer (US3):** shows only the trip name, organizer, dates and members, plus a notice that itinerary, availability and budget are Members-only. Buttons: "Accept invitation" and "Request to join" (then "Request sent"). Includes the invalid-link error.
4. **Trip overview (US3):** trip header (name, role badge, destination, dates, member count, Live indicator) and tabs Overview / Itinerary / Availability / Budget. Members list

(Organizer can remove members), join requests with Approve / Deny, share link with a Copy button, and trip settings (edit / delete; Members see "Leave trip").

5. **Itinerary (US4):** activities grouped by day in time order. Each activity shows time, title, location, who added it, who voted, a vote toggle with the count, and edit / delete. "Add activity" form (title, date, time required; location and notes optional). A live banner shows what another member just did; a "connection lost" state shows the fallback message.
6. **Availability (US5):** calendar of the trip dates; tapping a day marks the user free. Days get darker the more members are free and the best days get a BEST tag. Sidebar shows the range with the most overlap and who has not responded.
7. **Budget (US6):** summary (planned, spent, remaining), planned vs. actual bar per category with an "Over budget by $X" warning, member balances (+ is owed / − owes), expenses table, and a "Log expense" form (amount, category, payer, split among) showing the positive-two-decimals validation error.

Key user flows:

- New user: Sign up → Dashboard → New trip → Trip overview (becomes Organizer) → copy the share link.
- Invitee: open share link → Viewer page → Accept (becomes Member) or Request to join → Organizer approves on Trip overview.
- Planning: Itinerary → Add activity → others vote → everyone sees the change within about 2 seconds.
- Dates: Availability → mark free days → group picks the BEST range.
- Money: Budget → Log expense → category bars and balances update.

Section 6: Data model sketch

Users, Trips (including a share code), TripMemberships (user, trip, role, status of pending or active), ItineraryItems, Votes, Availability, Budgets, Expenses.

Section 7: Tech stack

Frontend: React + Tailwind CSS

- **Backend: Node.js + Express**
- **Database: MongoDB Atlas**
- **Auth: JWT in an httpOnly cookie, passwords hashed with bcrypt**
- **Live updates: Socket.IO**
- **Testing: Vitest (unit), Supertest (API integration), Playwright (E2E)**
- **CI/CD: GitHub Actions running tests + npm audit on every PR**
- **Hosting: Render**

Section 8: Issues

Issue	Title	Labels	Size	Sprint
Issue 1	Set up project framework	chore, priority: high	M	Sprint 1
Issue 2	Add AGENTS.md rules file for AI-assisted development	docs, priority: high	S	Sprint 1
Issue 3	Add user sign-up and login	feature, priority: high	M	Sprint 1
Issue 4	Create and view trips	feature, priority: high	M	Sprint 1
Issue 5	Generate trip share links and Viewer access	feature, priority: high	M	Sprint 1
Issue 6	Approve join requests and manage members	feature, priority: high	M	Sprint 1
Issue 7	Run tests automatically on every pull request	chore, priority: high	S	Sprint 1
Issue 8	Build shared itinerary of activities	feature, priority: high	M	Sprint 2
Issue 9	Add voting on itinerary activities	feature, priority: high	M	Sprint 2
Issue 10	Show live updates on the itinerary	feature, priority: high	M	Sprint 2
Issue 11	Schedule member availability	feature, priority: medium	L	Sprint 2
Issue 12	Track budget and expenses	feature, priority: medium	M	Sprint 2

Issue 13	Deploy app with automated deployment and PR previews	chore, priority: high	M	Sprint 2
Issue 14	Write API docs, diagrams and README	docs, priority: medium	M	Sprint 2

Created on GitHub with labels, milestones and acceptance-criteria checklists. PRD Issues 4, 5 and 9 were each split in two (they were the largest), and an issue for the AGENTS.md rules file was added for HW3, giving 14 issues.

Issue 1: Set up project framework

Labels: chore, priority:high · **Size:** M · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: Setup / infrastructure task (no user story).

Acceptance criteria:

- Frontend and backend folder structure is committed
- The app runs locally with one documented command
- A test runner is configured with one passing test

From PRD Issue 1.

Issue 2: Add AGENTS.md rules file for AI-assisted development

Labels: docs, priority:high · **Size:** S · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: Setup / infrastructure task (no user story).

Acceptance criteria:

- AGENTS.md exists in the repo root
- It covers tech stack and versions, folder structure, naming conventions and testing strategy
- It links the PRD and describes the mockups, key UI components and user flows
- It defines branch naming, commit message format and the PR workflow
- It lists do's and don'ts, preferred libraries, and security and accessibility rules

Required by HW3; both partners maintain it.

Issue 3: Add user sign-up and login

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: US1: As a new user, I want to create an account and log in so my trips and votes are linked to me.

Acceptance criteria:

- The backend connects to the MongoDB Atlas dev database through an environment variable
- A .env.example file is committed and .env is gitignored
- Registration takes name, email and password, and rejects duplicate emails
- Passwords are stored hashed
- Logged-in users stay logged in, and protected routes reject users who are not logged in
- Logout ends the session, and form errors are clear

From PRD Issue 2.

Issue 4: Create and view trips

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: US2: As a user, I want to create a trip so my group has one shared place to plan.

Acceptance criteria:

- A trip can be created with a name, destination and date range; the end date cannot be before the start date
- The creator becomes the Organizer
- The dashboard lists the user's trips, and the detail page shows Members
- The Organizer can edit or delete the trip

From PRD Issue 3.

Issue 5: Generate trip share links and Viewer access

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: US3: As an Organizer, I want to invite or approve Viewers so I control who joins.

Acceptance criteria:

- Each trip has a unique, random share code used in its link
- The Organizer and Members can copy the shareable link
- A logged-in user who opens a valid link sees the trip name and members as a Viewer
- Viewers cannot see the itinerary, availability or budget
- Invalid links show a clear error

First half of PRD Issue 4 (split because Issue 4 is size L).

Issue 6: Approve join requests and manage members

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: US3: As an Organizer, I want to invite or approve Viewers so I control who joins.

Acceptance criteria:

- A Viewer can accept an invitation or request to join

- The Organizer sees pending requests and can approve or deny each one
- The Organizer can remove Members
- Members can leave a trip

Second half of PRD Issue 4.

Issue 7: Run tests automatically on every pull request

Labels: chore, priority:high · **Size:** S · **Milestone:** Sprint 1 · **Board column:** Sprint Todo

User story: Setup / infrastructure task (no user story).

Acceptance criteria:

- A GitHub Actions workflow runs the test suite on every PR
- A failing test blocks the PR from showing a green check
- A dependency or security scan runs in CI

First half of PRD Issue 9, moved early so every Sprint 1 PR is tested.

Issue 8: Build shared itinerary of activities

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: US4: As a Member, I want a shared itinerary with voting and live updates so the group can decide, plan, and reserve together.

Acceptance criteria:

- Any Member can add an activity with a title, date and time, plus optional location and notes
- Members can edit or delete their own activities; the Organizer can edit or delete any activity
- Activities are grouped by day in chronological order
- Only Members can view the itinerary

First half of PRD Issue 5 (split because Issue 5 is size L).

Issue 9: Add voting on itinerary activities

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: US4: As a Member, I want a shared itinerary with voting and live updates so the group can decide, plan, and reserve together.

Acceptance criteria:

- Each Member can vote for an activity to show willingness to commit
- One vote per Member per activity, and it can be removed
- Each activity shows its vote count and who voted

Second half of PRD Issue 5.

Issue 10: Show live updates on the itinerary

Labels: feature, priority:high · **Size:** M · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: US4: As a Member, I want a shared itinerary with voting and live updates so the group can decide, plan, and reserve together.

Acceptance criteria:

- A new or edited activity, or a vote, by one Member appears on other Members' screens within about 2 seconds without a reload
- The connection status is shown
- If the live connection drops, the app shows a message and the page keeps working

From PRD Issue 7.

Issue 11: Schedule member availability

Labels: feature, priority:medium · **Size:** L · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: US5: As a Member, I want to mark my free dates so the group can find workable dates.

Acceptance criteria:

- Each Member selects free dates within the trip range and can update them
- The view highlights the dates with the most overlap
- The view shows which Members have not responded

From PRD Issue 6.

Issue 12: Track budget and expenses

Labels: feature, priority:medium · **Size:** M · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: US6: As a Member, I want to track spending against a budget so the trip stays affordable and fair.

Acceptance criteria:

- The Organizer sets a total budget, optionally broken down by category
- Members log expenses with amount, category, payer and who it is split among
- The summary shows planned vs. actual spending per category and in total, and flags overspending
- Each Member's balance is calculated
- Amounts must be positive and have two decimal places

From PRD Issue 8. If time runs short, simplify the balance calculation first.

Issue 13: Deploy app with automated deployment and PR previews

Labels: chore, priority:high · **Size:** M · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: Setup / infrastructure task (no user story).

Acceptance criteria:

- The deployed backend connects to the production database through environment variables set on the host
- The frontend and backend are deployed to a public URL
- Merging to main triggers deployment
- Pull requests get a deploy preview

Second half of PRD Issue 9; deploy previews are required by P2.

Issue 14: Write API docs, diagrams and README

Labels: docs, priority:medium · **Size:** M · **Milestone:** Sprint 2 · **Board column:** Backlog

User story: Setup / infrastructure task (no user story).

Acceptance criteria:

- OpenAPI documentation is published
- Architecture and database schema diagrams are committed
- The README explains setup and deployment

From PRD Issue 10.

Section 9: Sprint plan, risks, and open questions

Board: GitHub Project "TripVote Scrum Board" with columns Backlog → Sprint Todo → In Progress → In Review → Done.

Sprint 1 (milestone due Nov 5, 2026), issues 1, 2, 3, 4, 5, 6, 7: Set up project framework; Add AGENTS.md rules file for AI-assisted development; Add user sign-up and login; Create and view trips; Generate trip share links and Viewer access; Approve join requests and manage members; Run tests automatically on every pull request. These start in Sprint Todo.

Sprint 2 (milestone due Nov 12, 2026; P2 due Nov 13), issues 8, 9, 10, 11, 12, 13, 14: Build shared itinerary of activities; Add voting on itinerary activities; Show live updates on the itinerary; Schedule member availability; Track budget and expenses; Deploy app with automated deployment and PR previews; Write API docs, diagrams and README. These start in Backlog.

Risks: the largest features (invitations, itinerary + voting, availability) were split or scheduled so no single L-sized issue blocks a sprint. If time runs short, simplify the budget balance calculation first.